

AttestFlow

Proof-gated cross-chain payments for autonomous agents

Natural language for intent.
Attestcoin for truth.
Deterministic contracts for money.

3 linked public-testnet transactions · 62 live checks · 28 automated tests

AttestFlow

Source ↗

62 LIVE CHECKS PASSING

One payment. Two chains. No trusted payment oracle.

AttestFlow turns a natural-language settlement rule into a proof-bound CC3 action, then carries the exact on-chain result back to the source chain. Every hash below is public, final on testnet and re-verifiable.

The first use case is cross-border freelance escrow: the client pays where liquidity already exists, while Creditcoin releases only after the exact external payment is cryptographically proven.

ETHEREUM SEPOLIA01

A real client payment succeeds

0.01 native ETH · status 1 · payer and payee independently verified

0x6ac68ba9...59d555e7 ↗

CREDITCOIN CC302

The ASC proves, checks and settles

Attestcoin verify + proof-derived index + receipt status + replay guard

0xec29d5b4...89c2e4c2 ↗

ETHEREUM SEPOLIA03

The exact published payload executes

Same payload hash · authorized signature · destination replay guard

0xc692a176...796947fb ↗

SECURITY INVARIANTS

What the verifier proves

8 / 8

Proof inclusion	BlockProver verify() accepted on CC3
Index binding	calculateTxIndex() = Sepolia receipt index 69
Source success	Attested receipt status = 1
Policy match	Payee and 0.01 ETH value match the rule

INTEGRITY ANCHOR

Published payload hash

0x4845f5ca486987ddb30d486e58f36ed0cebbf5e514d20783d220b06f0d523faa

SourceTxId0x780a2c16...2f28fd1a

Released0.001 CTC

Destination0x83A0b8D2...194f868E

ATTESTFLOW · BUIDL CTC 2026 FALL

01

Autonomous payments have a truth problem

An AI agent can understand a payment rule.

It cannot be trusted to assert that the payment happened.

The usability layer may be probabilistic. The truth and money paths must be cryptographic and deterministic.

01 Centralized indexers

A service assertion becomes the settlement trigger.

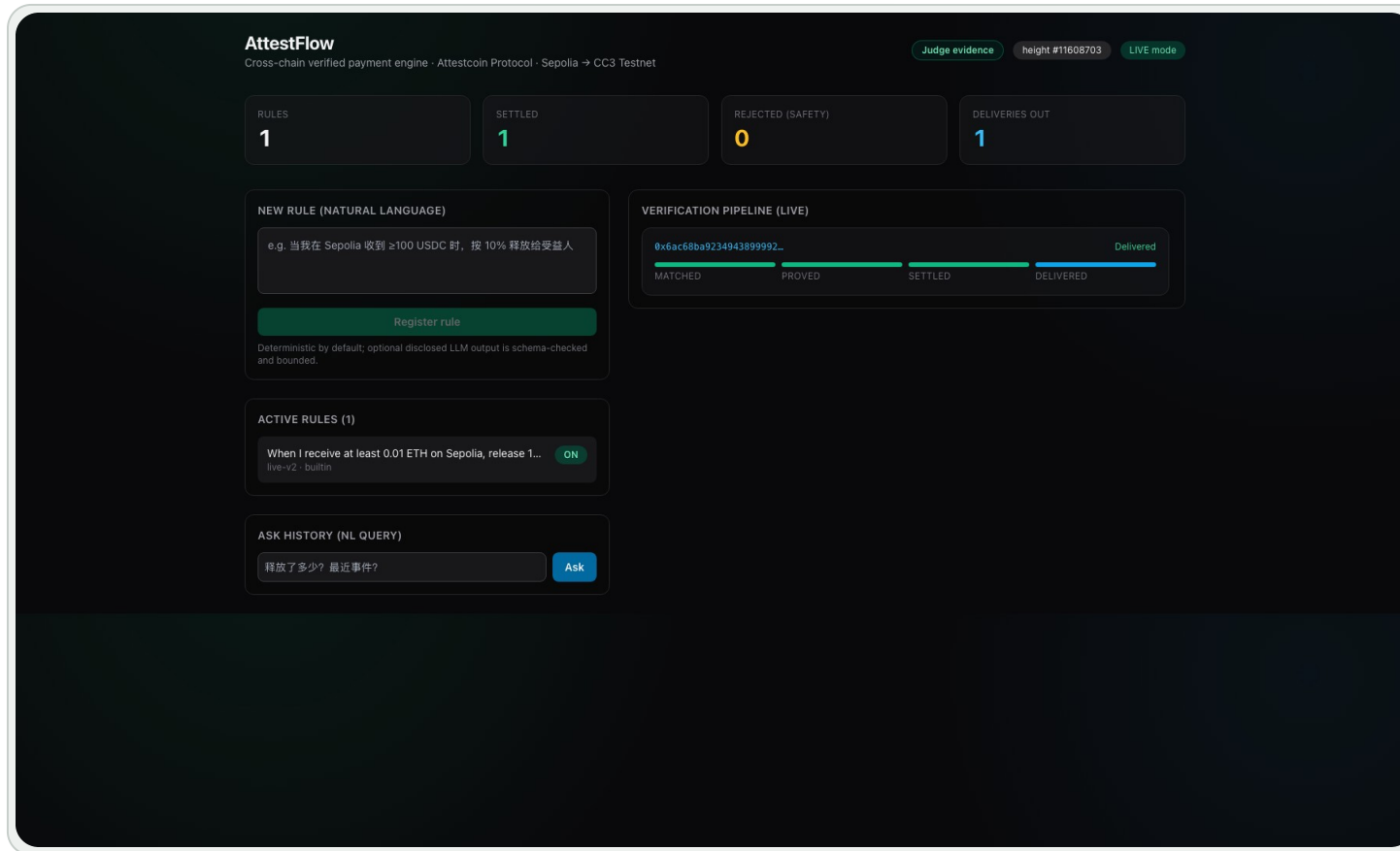
02 Bridge operators

Custody or relayer trust replaces source-chain truth.

03 Unchecked LLM output

A fluent answer can still be financially false.

Natural language never reaches money unchecked



1

Intent becomes policy

Exact amount, asset, payee, payout, beneficiary and destination.

2

Only attested blocks scan

Discovery cannot race ahead of Creditcoin attestation.

3

Proof gates settlement

The ASC—not the model—decides whether funds can move.

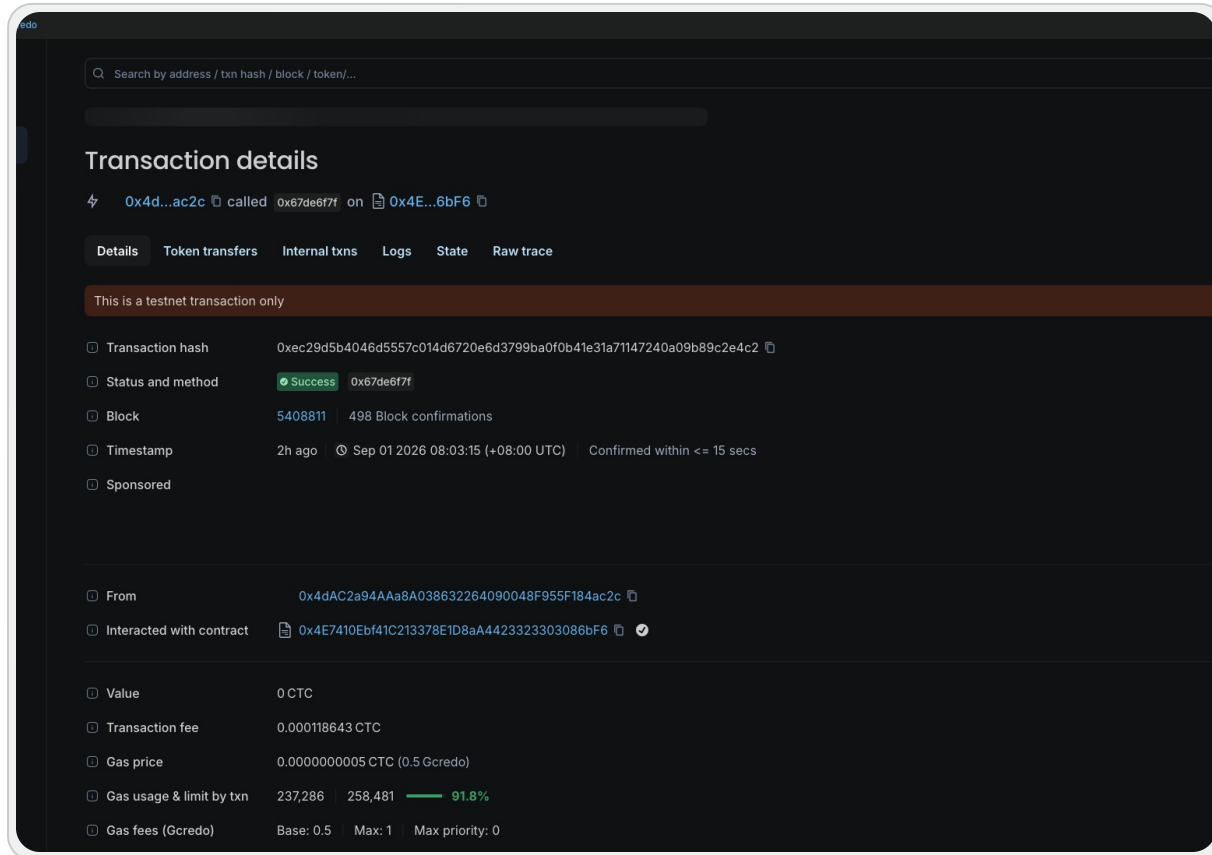
4

The exact result returns

Published bytes execute behind a second replay guard.

Optional LLM for parsing and Q&A. Mandatory deterministic validation at every financial boundary.

Attestcoin is the trust root—not a logo in the stack



01 ChainInfo · 0x0FD3

Sepolia chainId 11 155 111 must resolve to chainKey 1.

02 ProofBuilder

Builds the real Merkle and continuity proofs.

03 BlockProver · 0x0FD2

verify() synchronously gates CC3 state change.

04 calculateTxIndex()

Derives the replay identity from the Merkle path.

05 Encoding v1

Exposes transaction fields and receipt status to the ASC.

LIVE ASC 0x4E7410Eb41C213378E1D8aA4423323303086bF6

The ASC closes the gaps the precompile leaves open

ETHEREUM SEPOLIA 01

A real client payment succeeds

0.01 native ETH · status 1 · payer and payee independently verified

0x6ac68ba9_59d555e7 ↗

CREDITCOIN CC3 02

The ASC proves, checks and settles

Attestcoin verify + proof-derived index + receipt status + replay guard

0xec29d5b4_89c2e4c2 ↗

ETHEREUM SEPOLIA 03

The exact published payload executes

Same payload hash · authorized signature · destination replay guard

0xc692a176_796947fb ↗

SECURITY INVARIANTS 8 / 8

What the verifier proves

Proof inclusion	BlockProver verify() accepted on CC3
Index binding	calculateTxIndex() = Sepolia receipt index 69
Source success	Attested receipt status = 1
Policy match	Payee and 0.01 ETH value match the rule
Replay safety	ASC and Inbox replay guards are both set
Payload integrity	CC3 publish hash = Sepolia execute hash
Code identity	Deployed runtime bytecode = local Solidity build
Crash recovery	Destination leg resumed without a second settlement

INTEGRITY ANCHOR

Published payload hash

0x4845f5ca486987ddb30d486e58f36ed0cebbf5e514d20783d220b86f0d523faa

SourceTxId 0x780a2c16_2f28fd1a

Released 0.001 CTC

Destination 0x83A0b802_194f868E

HONEST BOUNDARY

Official Writability contracts are not live on this testnet. The return leg mirrors publish → sign → deliver → validate with one authorized relayer; it does not claim attester-quorum security.

REPRODUCE THE VERDICT

```

npm ci
npm run ci
npm run verify:evidence
# → { "status": "SUCCESS", "checks": 62 }
```

BLOCKPROVER PROVES

Transaction inclusion + source-chain continuity

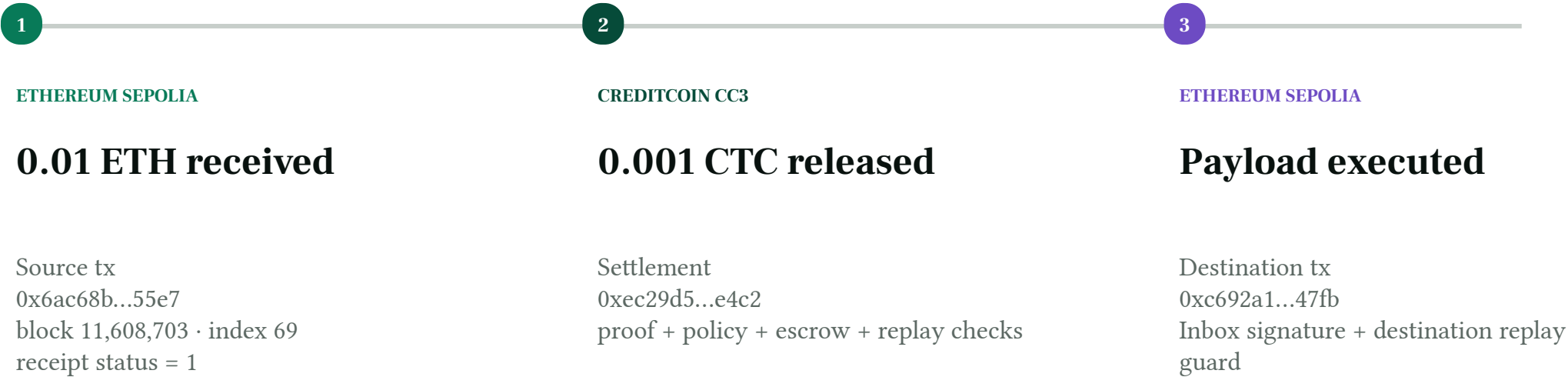
THE ASC ALSO ENFORCES

- Proof-derived transaction index
- Receipt status == 1
- Recipient · asset · calldata · amount
- Stored policy · beneficiary · destination
- Source replay guard · escrow solvency

A failed source transaction—or an operator-invented replay key—cannot settle.

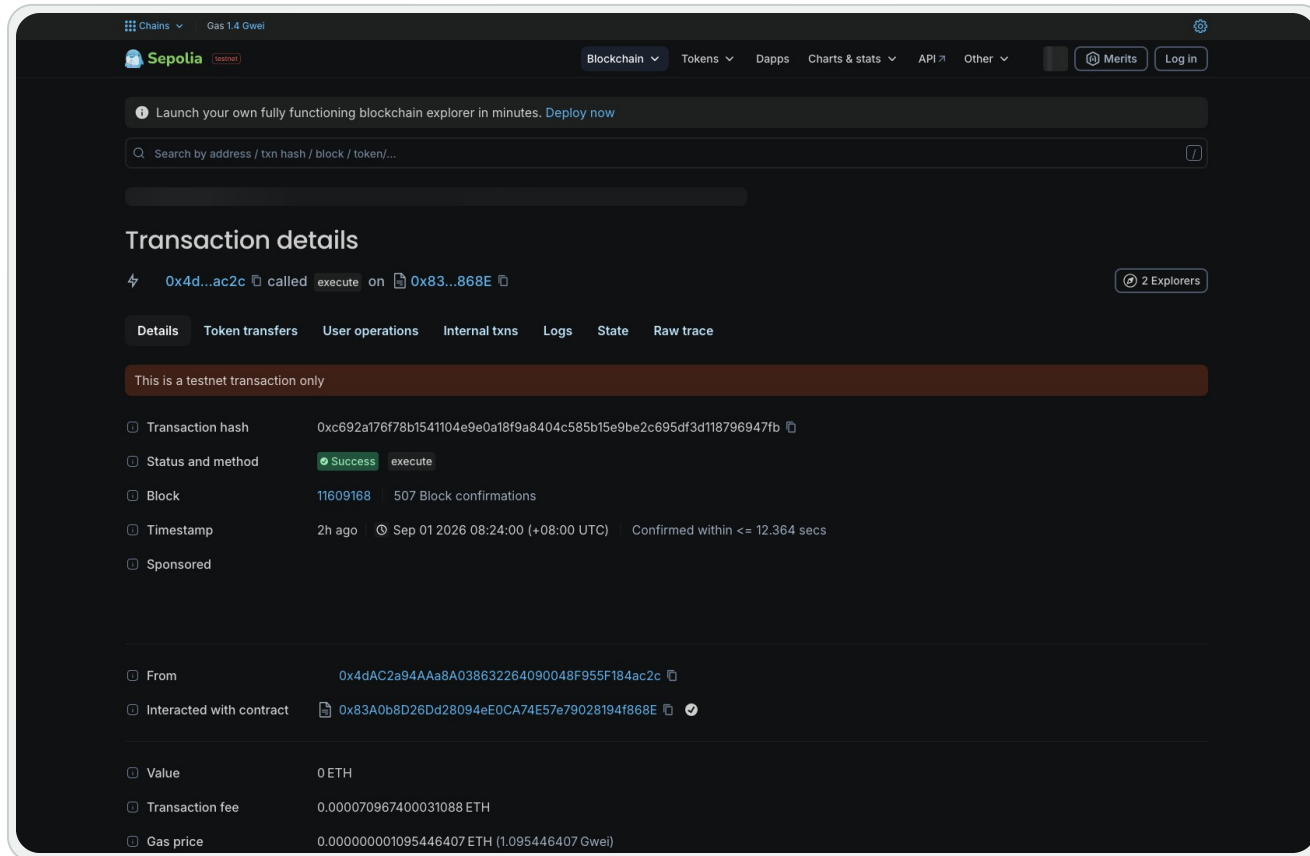
One payment. Three linked public-testnet legs.

The source payment, proof-gated settlement and destination execution are public—and tied by proof-derived identity plus one payload hash.



PAYLOAD HASH 0x4845f5ca486987ddb30d486e58f36ed0cebbf5e514d20783d220b06f0d523faa

The return leg is honest—and crash-resumable



CURRENT TESTNET BOUNDARY

Official Outbox/Inbox contracts are not deployed here. The adapter uses one authorized relay—*not* attester-quorum security.

RECOVERY EXERCISED IN THE LIVE RUN

- 01 CC3 settlement finalized
- 02 Sepolia RPC timed out
- 03 Restart recovered MessagePublished from chain logs
- 04 Only the missing destination leg completed

Published payload hash = executed payload hash

A judge can reproduce the verdict with one command

62

live cross-chain checks

28

automated tests

0

production vulnerabilities

Network identity · deployment provenance · source status · policy fields · settlement events · both replay guards · payload equality · deployed bytecode equality

attestflow — public-chain verification

```
$ npm run verify:evidence
```

```
{
  "status": "SUCCESS",
  "source": "0x6ac68b...55e7",
  "settlement": "0xec29d5...e4c2",
  "delivery": "0xc692a1...47fb",
  "sourceTxId": "0x780a2c...fd1a",
  "payloadHash": "0x4845f5...3faa",
  "checks": 62
}
```


Built for CEIP—not a one-off demo

Freelancer escrow is the wedge. The product is a reusable proof-gated payment engine for invoices, marketplaces, treasuries and agent-to-agent commerce.

NOW

Deployed testnet loop

Sepolia → Attestcoin → CC3 settlement →
Sepolia return

NEXT

Quorum-backed Writability

Swap the adapter for official Outbox/Inbox
when available

SCALE

Payment policy infrastructure

Multi-chain assets, invoice workflows and
agent commerce

Natural language for intent. Attestcoin for truth. Deterministic contracts for money.

github.com/ximilu0114-droid/ximilu-hackson